Chongzhou Fang

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Assistant Professor

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Education

- **University of California, Davis** — Davis, CA, USA
  PhD in Computer Engineering — *Sep. 2020 - Jun. 2025*
- **Southeast University** — Nanjing, China
  BSc in Information Science — *Aug. 2016 - Jun. 2020*

Research Interest

- System & System Security
- LLM for Hardware/Software Security
- IoT Firmware
- LLM Code Generation

Publications

[CCGRID'26] **Kumo: A Security-Focused Serverless Cloud Simulator**
*Wei Shao, Khaled Khasawneh, Setareh Rafatirad, Houman Homayoun and **Chongzhou Fang**.*

[ISQED'26] **Lightweight Heart Rate Variability Estimation Using PPG Signals**
*Ben Brown and **Chongzhou Fang**.*

[ISQED'26] **Fine-Grained Patching via Contention Balancer: A Practical Mitigation Strategy for Spectre Vulnerabilities Using SpecScope Contention Maps**
*Najmeh Nazari, Banafsheh Saber Latibari, Behnam Omidi, Hosein Mohammadi Makrani, Fatemeh Movafagh, Seyede Elahe Hosseini Imeni, **Chongzhou Fang**, Abhijit Mahalanobis, Khaled N Khasawneh, Houman Homayoun and Hossein Sayadi.*

[NDSS'26] **Bit of a Close Talker: A Practical Guide to Serverless Cloud Co-Location Attacks**
*Wei Shao, Najmeh Nazari, Behnam Omidi, Setareh Rafatirad, Khaled N Khasawneh, Houman Homayoun and **Chongzhou Fang**.*

[NDSS'26] **Know Me by My Pulse: Toward Practical Continuous Authentication on Wearable Devices via Wrist-Worn PPG**
*Wei Shao, Zequan Liang, Ruoyu Zhang, Ruijie Fang, Ning Miao, Ehsan Kourkchi, Setareh Rafatirad, Houman Homayoun and **Chongzhou Fang**.*

[ICCD'25] **FaRAccel: FPGA-Accelerated Defense Architecture for Efficient Bit-Flip Attack Resilience in Transformer Models**
*Najmeh Nazari, Banafsheh Saber Latibari, Elahe Hosseini, Fatemeh Movafagh, **Chongzhou Fang**, Hosein Mohammadi Makrani, Kevin Immanuel Gubbi, Abhijit Mahalanobis, Setareh Rafatirad, Hossein Sayadi and Houman Homayoun.*

[ACM TOIT] **Assessing and Mitigating Heterogeneity-Driven Security Threats in the Cloud**
***Chongzhou Fang**, Najmeh Nazari, Behnam Omidi, Han Wang, Aditya Puri, Manish Arora, Setareh Rafatirad, Houman Homayoun and Khaled Khasawneh.*

[ISCAS'25] **Extended Operational Life for Wearable Health Devices: A Hybrid TinyML and Server-Side ML Approach**
*Najmeh Nazari, Vedant Patel, **Chongzhou Fang**, Setareh Rafatirad and Houman Homayoun.*

[ISQED'25] **Navigating the trilemma: Security, power, and performance trade-offs in bluetooth low energy**
*Ning Miao, **Chongzhou Fang**, Ruijie Fang, Ruoyu Zhang, Mahdi Eslamimehr, Setareh Rafatirad, Hossein Sayadi and Houman Homayoun.*

[Book Chapter] **Cloud FPGA Accelerator Fingerprinting Using Communication Side**
Advances in Hardware Design for Security and Trust, CRC Press
***Chongzhou Fang**, Ning Miao, Han Wang, Jiacheng Zhou, Tyler Sheaves, John M Emmert, Avesta Sasan and Houman Homayoun.*

[JMIR AI] **Survey on pain detection using machine learning models: narrative review**
*Ruijie Fang, Elahe Hosseini, Ruoyu Zhang, **Chongzhou Fang**, Setareh Rafatirad and Houman Homayoun.*

[IEEE Access] **Transformers: A security perspective**
*Banafsheh Saber Latibari, Najmeh Nazari, Muhtasim Alam Chowdhury, Kevin Immanuel Gubbi, **Chongzhou Fang**, Sujan Ghimire, Elahe Hosseini, Hossein Sayadi, Houman Homayoun, Soheil Salehi and Avesta Sasan.*

[BSN'24] **Validation of webe band during physical activities**
*Ruijie Fang, Sally Hang, Ruoyu Zhang, **Chongzhou Fang**, Setareh Rafatirad, Camelia Hostinar and Houman Homayoun.*

[BSN'24] **Advanced energy-efficient system for precision electrodermal activity monitoring in stress detection**
*Ruoyu Zhang, Ruijie Fang, Elahe Hosseini, **Chongzhou Fang**, Ning Miao and Houman Homayoun.*

[UsenixSecurity'24a] **Large Language Models for Code Analysis: Do LLMs Really Do Their Job?**
***Chongzhou Fang**, Ning Miao, Shaurya Srivastav, Jialin Liu, Ruoyu Zhang, Ruijie Fang, Asmita Asmita, Ryan Tsang, Najmeh Nazari, Han Wang and Houman Homayoun.*

[UsenixSecurity'24b] **Forget and Rewire: Enhancing the Resilience of Transformer-based Models against Bit-Flip Attacks**
*Najmeh Nazari, Hosein Mohammadi Makrani, **Chongzhou Fang**, Hossein Sayadi, Setareh Rafatirad, Khaled N. Khasawneh and Houman Homayoun.*

[UsenixSecurity'24c] **Fuzzing BusyBox: Leveraging LLM and Crash Reuse for Embedded Bug Unearthing**
*Asmita Asmita, Yaroslav Oliinyk, Michael Scott, Ryan Tsang, **Chongzhou Fang** and Houman Homayoun.*

[DAC'24] **Architectural Whispers: Unveiling Machine Learning Models with Frequency Throttling Side-Channel Fingerprinting**
*Najmeh Nazari, **Chongzhou Fang**, Hosein Mohammadi Makrani, Behnam Omidi, Setareh Rafatirad, Avesta Sasan, Hossein Sayadi, Houman Homayoun and Khaled N. Khasawneh.*

[ISQED'24] **LLM-FIN: Large Language Models Fingerprinting Attack on Edge Devices**
*Najmeh Nazari, Furi Xiang, **Chongzhou Fang**, Hosein Mohammadi Makrani, Aditya Puri, Kartik Patwari, Hossein Sayadi, Setareh Rafatirad, Chen-Nee Chuah and Houman Homayoun.*

[ISCAS'24] **Securing On-Chip Learning: Navigating Vulnerabilities and Potential Safeguards in Spiking Neural Network Architectures**
*Najmeh Nazari, Kevin Immanuel Gubbi, Banafsheh Saber Latibari, Muhtasim Alam Chowdhury, **Chongzhou Fang**, Avesta Sasan, Setareh Rafatirad, Houman Homayoun and Soheil Salehi.*

[DATE'24] **SpecScope: Automating Discovery of Exploitable Spectre Gadgets on Black-Box Microarchitectures**
*Najmeh Nazari, Behnam Omidi, **Chongzhou Fang**, Hosein Mohammadi Makrani, Setareh Rafatirad, Avesta Sasan, Houman Homayoun and Khaled N. Khasawneh.*

[BIBM'23] **Introducing an Open-Source Python Toolkit for Machine Learning Research in Physiological Signal based Affective Computing**
*Ruijie Fang, Ruoyu Zhang, Elahe Hosseini, **Chongzhou Fang**, Setareh Rafatirad and Houman Homayoun.*

[CCS'23]	**Gotcha! I Know What You are Doing on the FPGA Cloud: Fingerprinting Co-Located Cloud FPGA Accelerators via Measuring Communication Links**
(CSAW'24 ARC Tech Impact Award Runner-Up)	***Chongzhou Fang**, Ning Miao, Han Wang, Jiacheng Zhou, Tyler Sheaves, John M Emmert, Avesta Sasan and Houman Homayoun.*
[ICCAD'23]	**Side Channel-Assisted Inference Attacks on Machine Learning-Based ECG Classification**
	*Jialin Liu, Houman Homayoun, **Chongzhou Fang**, Ning Miao and Han Wang.*
[Ubicomp/ISWC'23 Adj.]	**Privee: A Wearable for Real-Time Bladder Monitoring System**
	*Ruoyu Zhang, Ruijie Fang, **Chongzhou Fang**, Houman Homayoun and Gozde Goncu Berk.*
[CODES+ISSS'23]	**Special Session: Mitigating Side-channel Attacks through Circuit to Application Layer Approaches**
	*Nima Kavand, Armin Darjani, Jens Trommer, Giulio Galderisi, Thomas Mikolajick, Nicolai Müller, Amir Moradi, **Chongzhou Fang**, Ning Miao, Han Wang, Sai Manoj Pudukotai Dinakarrao, Houman Homayoun, Benjamin Hettwer, Luca Parrini and Akash Kumar.*
[DAC'23]	**Don't Cross Me! Cross-Layer System Security**
	*Najmeh Nazari, **Chongzhou Fang**, Sai Manoj PD, Houman Homayoun.*
[IEEE Micro]	**Adversarial Attacks Against Machine Learning-Based Resource Provisioning Systems**
	*Najmeh Nazari, Hosein Mohammadi Makrani, **Chongzhou Fang**, Behnam Omidi, Setareh Rafatirad, Hossein Sayadi, Khaled N. Khasawneh and Houman Homayoun.*
[NDSS'23]	**HeteroScore: Evaluating and Mitigating Cloud Security Threats Brought by Heterogeneity**
	***Chongzhou Fang**, Najmeh Nazari, Behnam Omidi, Han Wang, Aditya Puri, Manish Arora, Setareh Rafatirad, Houman Homayoun and Khaled N. Khasawneh.*
[NDSS'22]	**Repttack: Exploiting Cloud Schedulers to Guide Co-Location Attacks**
	***Chongzhou Fang**, Han Wang, Najmeh Nazari, Behnam Omidi, Avesta Sasan, Khaled N. Khasawneh, Setareh Rafatirad and Houman Homayoun.*

Teaching and Mentoring

- **CMPE-250 Assembly and Embedded Programming** — RIT
  Instructor — *Spring 2026*
- **CMPE-789 Special Topics: Modern Cloud Computing** — RIT
  Instructor — *Fall 2026*

Awards

- RIT seed funding. ($10,000)
- CSAW Applied Research Competition Finalist (15 out of 194 submissions) & Technical Impact Award Runner-Up, 2024.

Services

- **Organizer:**
  - LLM-Assisted Security and Trust Exploration (LAST-X) Workshop, 2026 & 2027. (Co-located with NDSS)
  - LLMs for Hardware Design Challenge, 2025. [Co-located with IEEE International Conference on Computer Design (ICCD 2025)]
- **Technical Program Committee:**
  - IEEE Annual Computer Security Applications Conference (ACSAC) 2026.

  - Network and Distributed System Security Symposium (NDSS) 2026 & 2027.
  - IEEE International Symposium on Hardware Oriented Security and Trust (HOST) 2026.
  - Great Lakes Symposium on VLSI (GLSVLSI) 2026, IoT and Smart System Track.
- **Journal Reviewer:**
  - IEEE Transactions on Dependable and Secure Computing (TDSC).
  - IEEE Transactions on Information Forensics and Security (TIFS).
  - ACM Transactions on Internet of Things (TIOT).
  - ACM AI Letters.